

# MRV Stack: White-Box, Auditor-Compliant Measurement

## Executive Summary

The Measurement, Reporting, and Verification (MRV) methodology is the single most important component of any ERW carbon credit operation. If the auditor cannot verify your carbon claims, you do not get paid. We have developed a deterministic, physics-based MRV stack that is fundamentally different from the industry's prevailing approach.

## The Industry's Current Approach (And Why It Fails)

### What Most Companies Do

1. Collect soil samples before and after basalt application
2. Send samples to expensive third-party labs for cation analysis
3. Feed results into a Deep Learning / machine learning model
4. Use the model's output to claim carbon credits

### Why This Fails Audits

- **Lab costs are prohibitive at scale:** At 50–200 per sample, monitoring thousands of hectares becomes economically unviable
- **ML models are black boxes:** Carbon registries (Puro.earth, Isometric) increasingly require deterministic, physically interpretable methodologies
- **Temporal resolution is too low:** Quarterly or annual lab samples miss the critical “Year 1 Flush” dynamics entirely
- **No physical mass-balance check:** ML models can predict any number; without a physics constraint, there is no way to verify the prediction is physically possible

## Our Approach: The White-Box MRV Stack

### Layer 1: Continuous EC Sensor Network

- Deploy low-cost Electrical Conductivity sensors at 30 cm depth across the deployment zone

- EC is a direct, real-time proxy for total dissolved ion concentration in pore water
- Provides continuous (hourly/daily) temporal resolution vs. quarterly lab visits
- Cost: ~15–30 per sensor vs. 50–200 per lab sample

## **Layer 2: Kohlrausch's Law Calibration**

- Kohlrausch's Law provides a deterministic, linear relationship between EC and individual ion concentrations
- Calibrate the EC-to-ion relationship using a small number of initial lab samples
- Once calibrated, the EC sensors provide continuous, real-time estimates of  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ , and total alkalinity flux
- This is a published, peer-reviewed physical law — not a trained model

## **Layer 3: Isotopic Verification ( $^{87}\text{Sr}/^{86}\text{Sr}$ )**

- Strontium isotope ratios provide a definitive “fingerprint” distinguishing basalt-derived ions from background soil weathering
- Periodic isotopic sampling (quarterly) provides an independent cross-check on the EC-derived estimates
- This is the gold standard for source attribution in geochemistry

## **Layer 4: Mass-Balance Physical Audit**

- Total ion export (from EC sensors) is constrained by the physical mass of basalt applied
- If the model predicts more carbon capture than is physically possible given the mass of rock deployed, the estimate is automatically flagged and capped
- This hard physical constraint is what separates our approach from unconstrained ML predictions

## Cost Comparison

| Component                   | Industry Standard       | Our Stack                | Savings             |
|-----------------------------|-------------------------|--------------------------|---------------------|
| Per-hectare monitoring cost | 150–400/ha/year         | 30–80/ha/year            | ~80% reduction      |
| Temporal resolution         | Quarterly               | Continuous (daily)       | 90× improvement     |
| Auditability                | Black-box ML            | Deterministic physics    | Registry-compliant  |
| Scalability                 | Limited by lab capacity | Unlimited (sensor-based) | Orders of magnitude |

## What We Are Not Sharing Here

- The exact Kohlrausch calibration coefficients for different soil types
- Our sensor deployment protocol and optimal placement geometry
- The proprietary algorithm linking EC flux to carbon credit issuance timing
- The specific registry submission templates we have developed